6.X Connection to the Full Relativity (FR) framework

The Quaia redshift–dipole analysis sits naturally inside the broader *Full Relativity* (FR) framework which replaces an expanding metric spacetime with a variable speed of light and evolving mass-per-unit-matter in a non-expanding universe.

Within FR, the cosmological redshift is interpreted as a consequence of lower stored energy per unit matter at earlier epochs, rather than stretching of photon wavelengths by an expanding "space".

The DES and Union 2.1 Type Ia supernova data, once distances are divided by $\left(\frac{1}{2}+z\right)$ to correct for the higher light speed at emission, yield an almost perfectly linear time–redshift relation with slope ≈ 14.47 Gyr per unit z.

This corresponds to a Hubble constant $H_0 \simeq 67.6$ km s^{-1} Mpc^{-1} is in excellent agreement with early-time (Planck) constraints, without invoking dark energy or an accelerating expansion phase.

Within FR, the apparent "Hubble tension" arises as a modelling artefact: late-time analyses such as SH0ES effectively apply an expansion-history correction to the redshift (through the deceleration parameter q_0) on top of a data set whose brightness–redshift trend is already consistent with a constant-slope, non-expanding interpretation.

When those same DES-derived trends are multiplied by $\left(\frac{1}{2}+z\right)$ and then run through the SH0ES expansion-history correction, one recovers a high value $H_0 \approx 73$ km s^{-1} Mpc^{-1} closely matching the Pantheon+/SH0ES result.

In FR this is understood as double-counting: the same physical effect (higher light speed and lower mass per unit matter at emission) is being encoded once in the supernova distances and again as an assumed expansion history.

The CMB dipole "anomaly" is also reinterpreted in this framework. Under FR, the CMB spectrum reflects a much lower energy scale at the time when hydrogen could first form and photons decouple, not a hot big bang plasma cooled by expansion.

The observed CMB dipole suggests a straightforward kinematic interpretation:
Doppler shifting of this nearly uniform background by our motion at ≈ 370 km/s relative to a stationary background field.

The Quaia analysis, using mean quasar redshifts binned on the sky with a $/b/ > 20^{\circ}$ Galactic mask, finds that the *low-z* ($z \lesssim 1$) redshift dipole aligns in direction with the CMB dipole and that this alignment is robust to randoms, shuffles, regressions, jackknife cuts, and alternative masks.

This supports the view that both the CMB and the low-z galaxy/quasar redshift fields trace the same kinematic motion relative to a non-expanding background, rather than requiring an additional "intrinsic" CMB dipole or subtle systematics in the number–count dipole alone.

More broadly, FR offer an alternative to GR's curvature of a malleable spacetime with changes in light speed, inertia, and mass per unit matter that depend on the surrounding matter–antimatter distribution.

This naturally predicts higher inertia in galaxy interiors and around black holes, offering an avenue to account for flat rotation curves, gravitational lensing, and the Bullet Cluster dynamics without invoking dark matter, while also eliminating the need for dark energy once the supernova data are re-analysed under a variable-c, non-expanding cosmology.